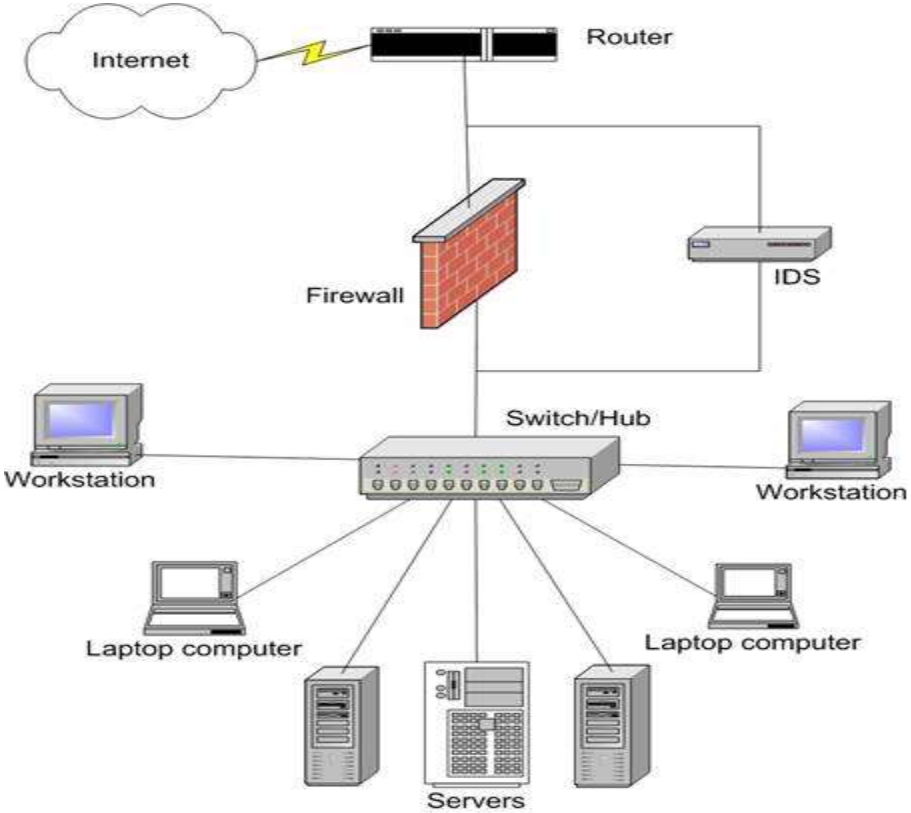
# Foundation of Networking.

1.What is Networking?

A collection devices that can communicate together.



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- 2.Pieces of Network.
  - Ethernet cable.
  - Switch
  -Router
  -Computer
  -Server
  -NIC (Network interface card)
  -Wireless Ethernet card
  -Service Provider.

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- 3.Application that use the network.

  -Web browser

  -Database application

  -Instant Messenger

  -Email

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- 4. Consideration for Network Application.

-Speed

*Bit>byte/>Kilobyte>Megabyte>Gigabyte>Terabyte and Exabyte. For internet speed we measure in kilobit per second(Kbps) , Megabit per second (Mbps), Gigabit per second (Gbps) etc.*

-Delay

VoIP

-Availability

.

- Network Topology/Design.

  -Bus, Ring, Star, Hybrid and Mesh.

# Open Systems Interconnection Model (OSI Model).

- It helps to break down network functions.
- Creates standards for equipment manufacturing.
- Allows vendors to focus in specialized areas.

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**OSI Model Layers.**

-Application layer –Provides network access to the application e.g e-mail, web etc.

-Presentation Layer – Generifies the data – Encryption services.

-Session – Starts and ends the session – Logically keeps the session separate.

.

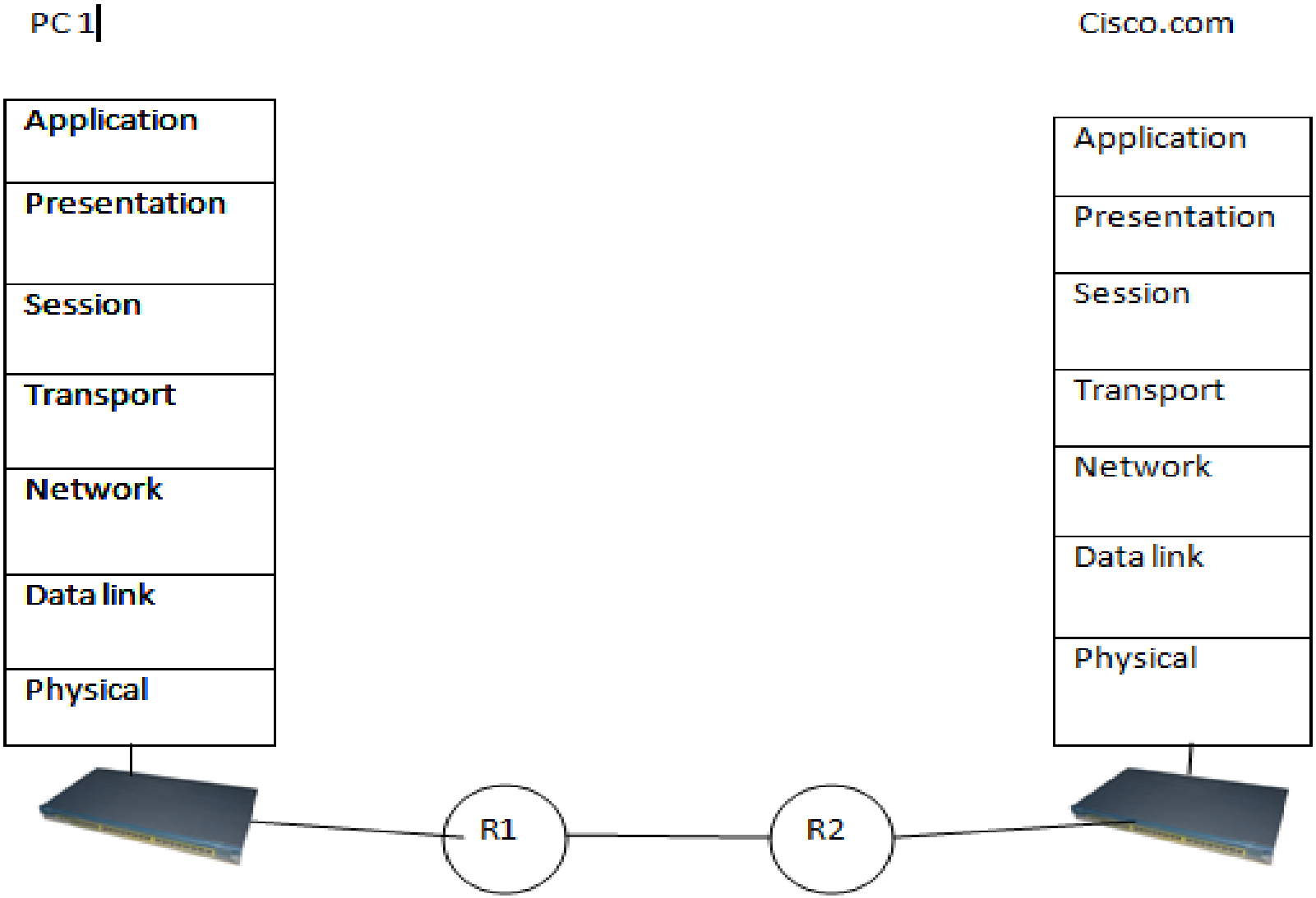
-Transport Layer – Dictates how the data is sent-TCP/UDP and defines well known ports.

-Network Layer – Provides logical addressing, find the best path to destination.

-Datalink Layer - Physical address, ensure data is errofree. Switches.

-Physical Layer – Network cable-UTP, Serial cable, coaxial cable and Fibre Optic cable.

# OSI Model in real world.



# How a PC 1 access the cisco.com

- Through a web browser at the application layer PC1 with IP address of 10.10.1.15 will try to access cisco.com DNS server with IP address of 66.48.5.8.
- At the presentation layer the format of data will be checked. Data encryption will be checked i.e http or https. In our case http will be used
- At the session layer , the session will me started and managed by separating it from other session.

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- At the transport layer the TCP session will be started. Request and ACK as well as the source and destination port number.
- Port 80 is for web server. Source can be any port e.g 1020. *ftp* *ftp.microsoft.com* *and netstat*

- At the Network layer, the Destination and surce ip address will be checked in our case is 66.48.5.8 and 10.10.1.15.

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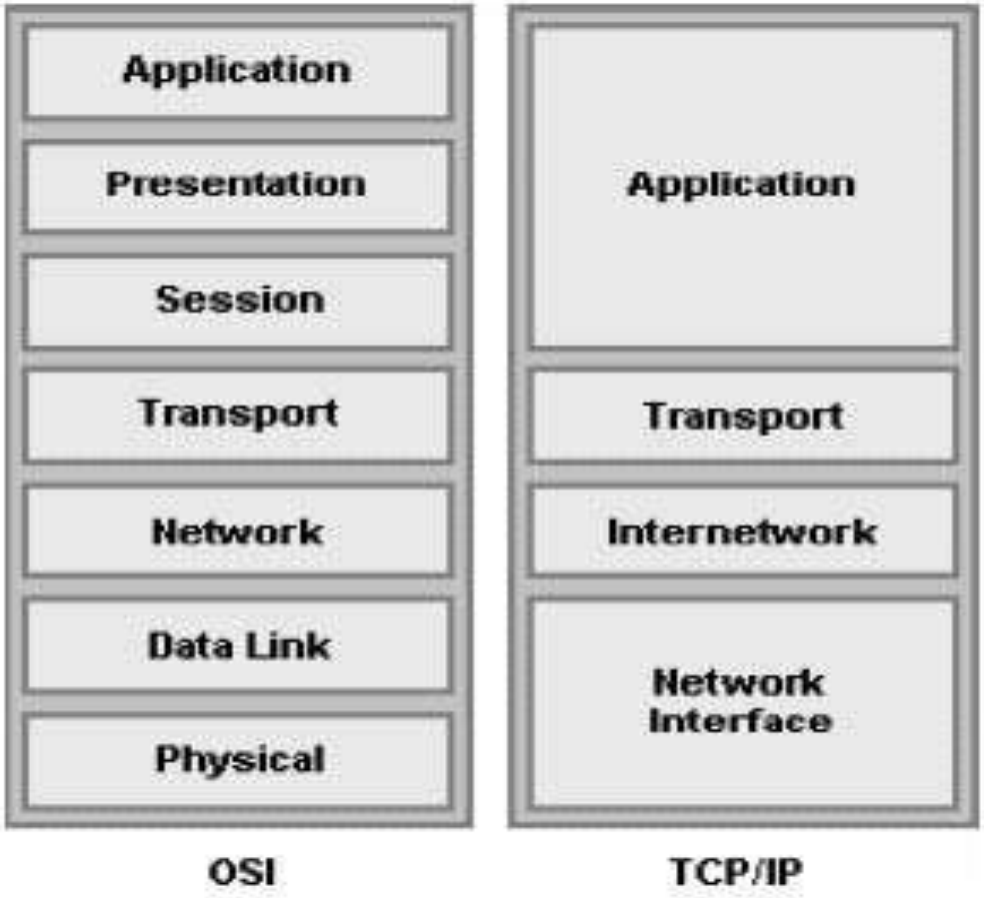
- At the Data link that where the Destination and source MAC address will be exchanged. Note that the MAC addresses will change end to end where as the IP address will never change. The first destination MAC address will be the MAC address of the R1,Because of the IP address in the packet the router forward the frame. Source MAC is MAC address of the PC.
- At a physical layer it will pass the wire to the Physical layer (NIC) of the cisco server. Check whether is its MAC address and send it layer by layer to the application layer.

# Basic TCP/IP: Addressing Fundamentals



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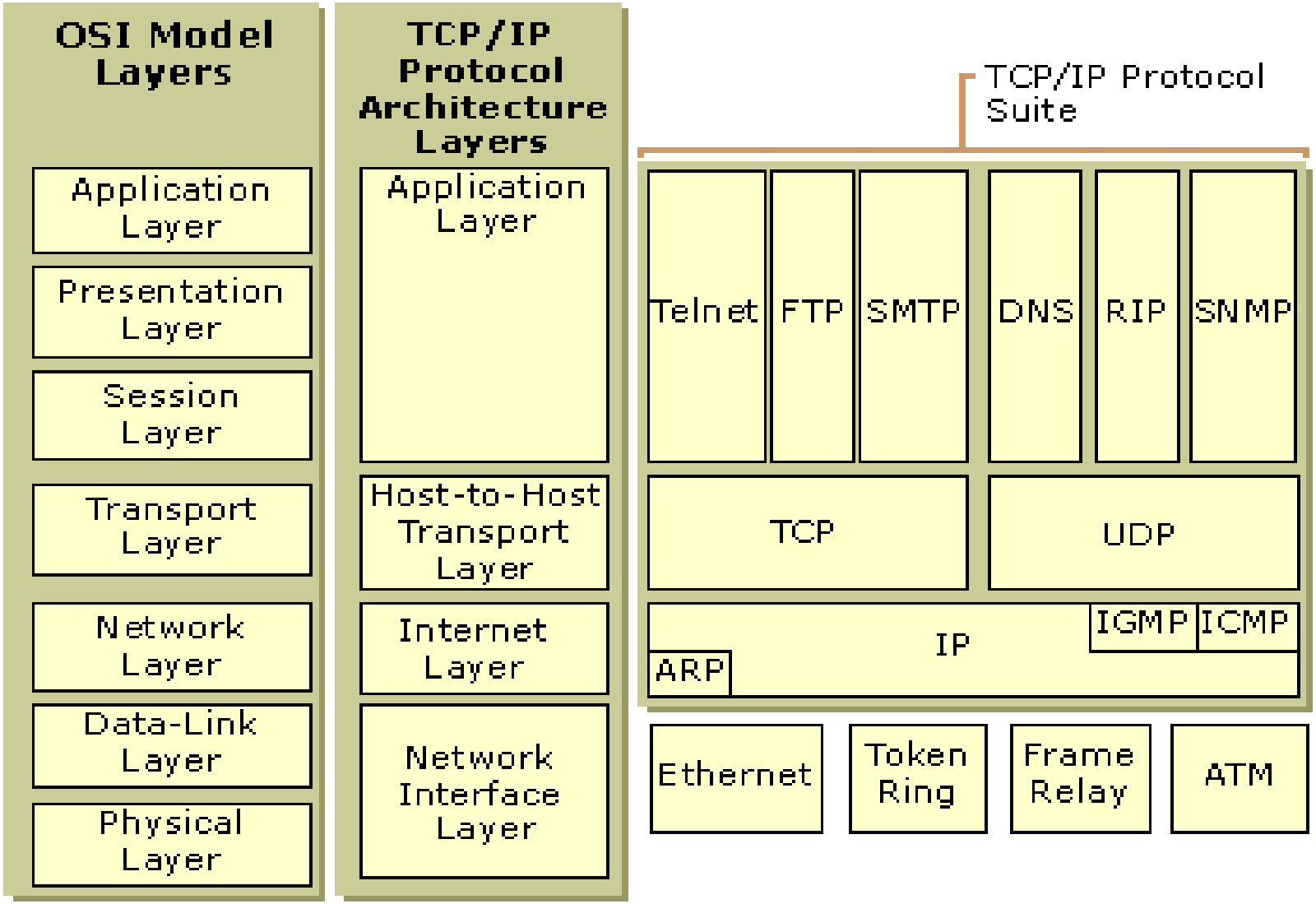
- **How OSI and TCP/IP Model relates**.

-The OSI model is used to describe how network communicate while TCP/IP model used to describe how the communication happen.

-TCP/IP was created by the Department of Defense (DoD) to ensure and preserve data integrity, as well as maintain communications in the event of catastrophic war.

-TCP/IP is a suites of protocols such as Telnet, FTP, SMTP, TCP, UDP, ICMP, IP, Frame relay and Ethernet just to mention few.


OSI Model Layers
Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data-Link Layer
Physical Layer
TCP/IP Protocol Architecture Layers
Application Layer
Host-to-Host Transport Layer
Internet Layer
Network Interface Layer
TCP/IP Protocol Suite
Telnet
FTP
SMTP
DNS
RIP
SNMP
TCP
UDP
IP
IGMP
ICMP
ARP
Ethernet
Token Ring
Frame Relay
ATM

.

- **IP Address format.**

It has four numbers, each number from 0-255

Example of IP address: 172.30.5.82.

IP address normally combines subnet mask and default gateway.

IP: 172.30.5.82

Mask:255.255.0.0

Gateway:172.30.5.1

Note: The subnet mask shows which portion of the IP address identifies the network and host.

# Communication between same network

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- Referring to above network, when PC with the IP address 172.30.5.82 tries to communicate with PC with IP address 172.30.5.83, ARP broadcast will be sent to all interface connected to the same LAN including the router interface asking who is 172.30.5.83?

172.30.5.83 will respond with its MAC address and the communication will start.

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- ARP (Address Resolution Protocol) is a crucial network protocol that translates IP addresses to MAC addresses within a local network.
- It works by allowing devices on a network to discover the MAC address of a specific IP address, which is necessary for sending data packets at the data link layer.

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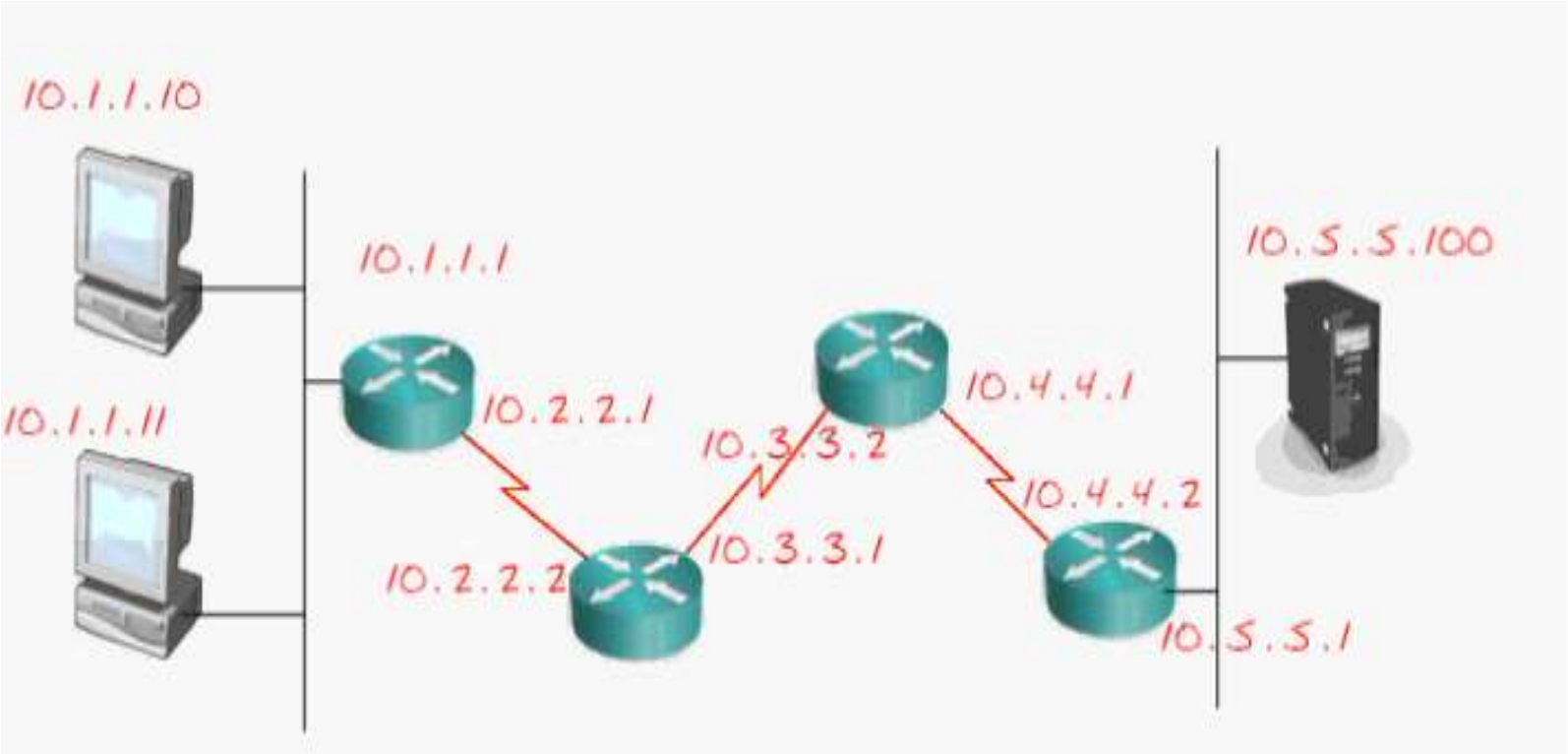
- The device that sent the ARP request receives the reply and stores the mapping between the IP and MAC addresses in its ***ARP cache*** (a local table).
- This allows the source device to send future data packets directly to the destination device's MAC address, without needing to broadcast ARP requests again for the same IP address.

# Communication between different networks



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- Referring to above network, let us assume PC 1 with IP 10.1.1.10 wants to communicate with Server with IP address 10.5.5.100. It will send ARP broadcast to the router asking who is 10.5.5.100.
- The source MAC address will be PC1 MAC address and the destination MAC address will be the MAC address of the router. Since the destination IP address is not belong to the router the router will forward the packet to its destination.

.

- The source MAC address will be the R1 MAC address and the destination MAC address will be of R2.It will go like that until it reaches the Server .
- The server will respond with the source IP address of 10.5.5.100 and destination IP address of 10.1.1.10.

# Usable class addresses.

There are 3 usable class addresses i.e Class A, Class B and Class C.

Class A:

- First octet of IP address is 1-126
- Example: 10.5.1.1
- Has the subnet mask of 255.0.0.0 i.e NHHH.

For class A you will get the total number of 16777214 hosts,however Cisco recommends not to exceed 500 hosts on a single network.

Class B:

- First octet of ip address is 128-191
- Example 150.51.233.1
- Has the subnet mask 255.255.0.0 i.e NNHH It has 65534 hosts.

Class C:

- First octet of ip address is 192-223.
- Example: 220.1.50.63
- Has a subnet mask of 255.255.255.0 i.e NNNH It has 254 hosts.

# Public and Private IP addresses:

- **Public addresses:**

-Usable only on internet and external networks.

. **Private addresses:**

**-**Usable only on internal networks.

**Three ranges of Private IP addresses.**

Class A: 10.0.0.0 -10.255.255.255

Class B: 172.16.0.0 – 172.31.255.255

Class C: 192.168.0.0 – 192.168.255.255.

-Loop back range used for testing 127.0.0.0

-Auto-configuration range 169.254.X.X.

# TCP and UDP Communication

- **TCP:**

-It is connection-oriented.

-Uses sequence numbers.

-Reliable (uses ACKS) i.e. the server will send an ACK that it received it.

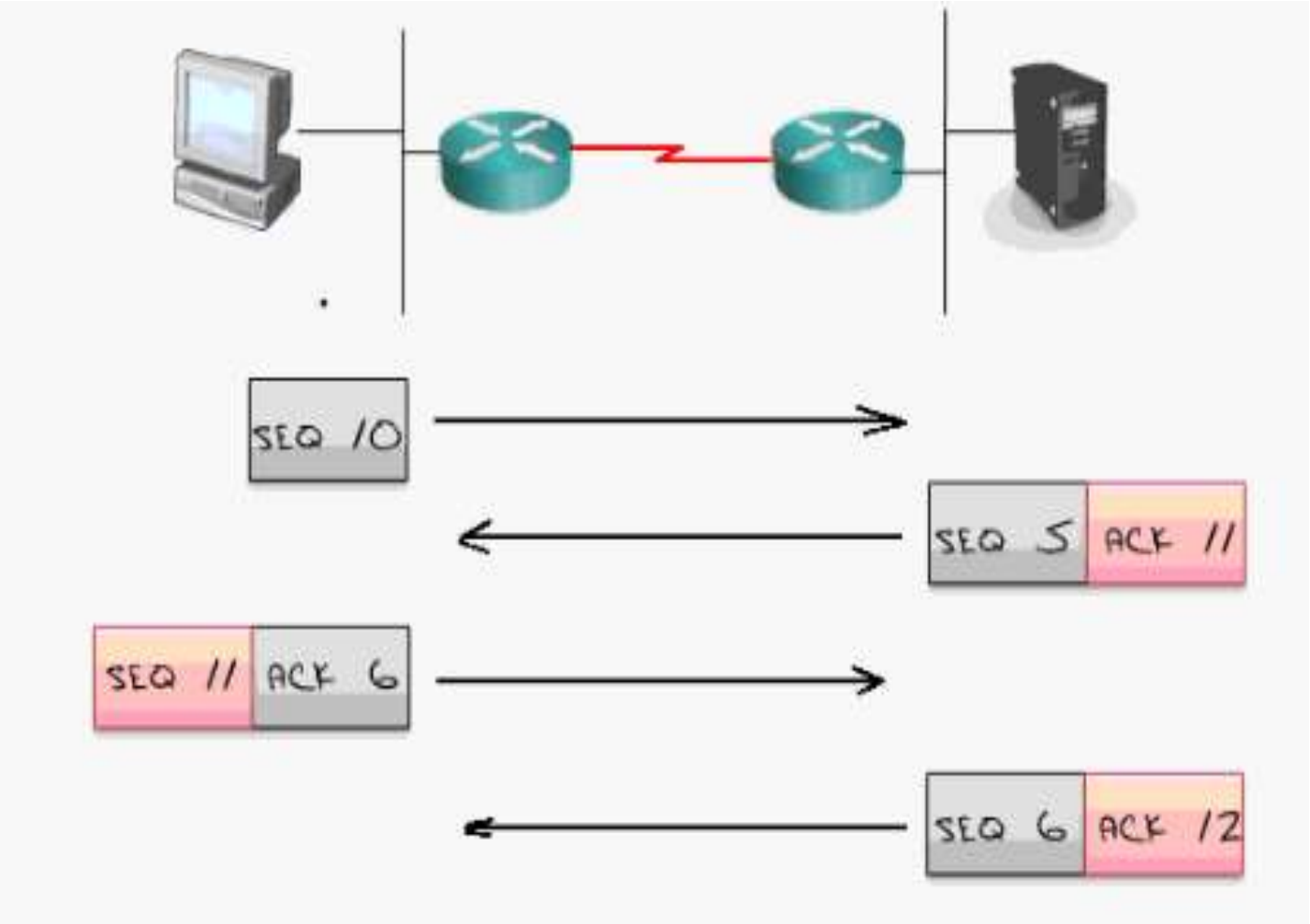
*Three way-hand shake*.

SYN →

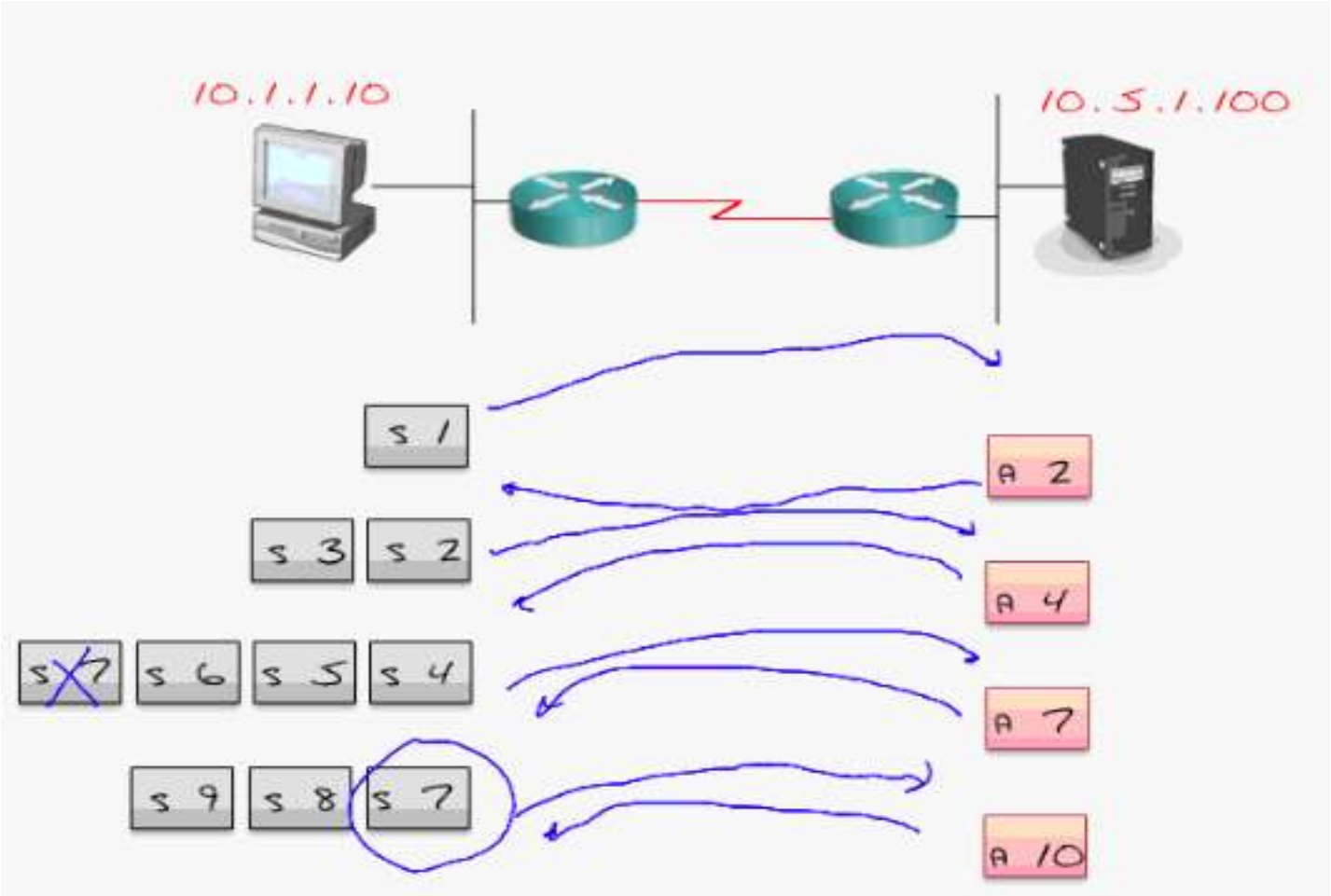
← SYN-ACK

ACK →

# -Sequence numbers:



# -TCP Windowing



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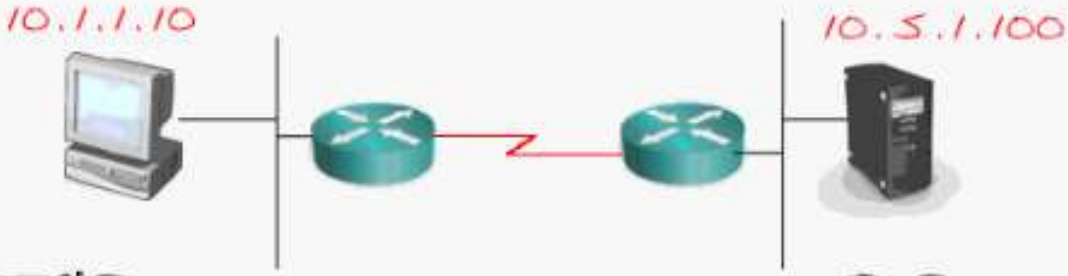
In TCP Windowing the computer will increase the amount of data to be send depending how reliable the connection is. As long as the other side sends the ACK back the window will be increased sequentially until some packets are droped. Once this happens the computer will reduce the amount of packets.

# Port numbers.




TCP:
<0-65535>

UDP:
<0-65535>

21 - FTP
22 - SSH
23 - TELNET
25 - SMTP
53 - DNS SERVER
80 - HTTP
110 - POP3
443 - HTTPS

53 - DNS CLIENT
69 - TFTP

NOTE: 0-1023 ARE CONSIDERED WELL-KNOWN PORTS

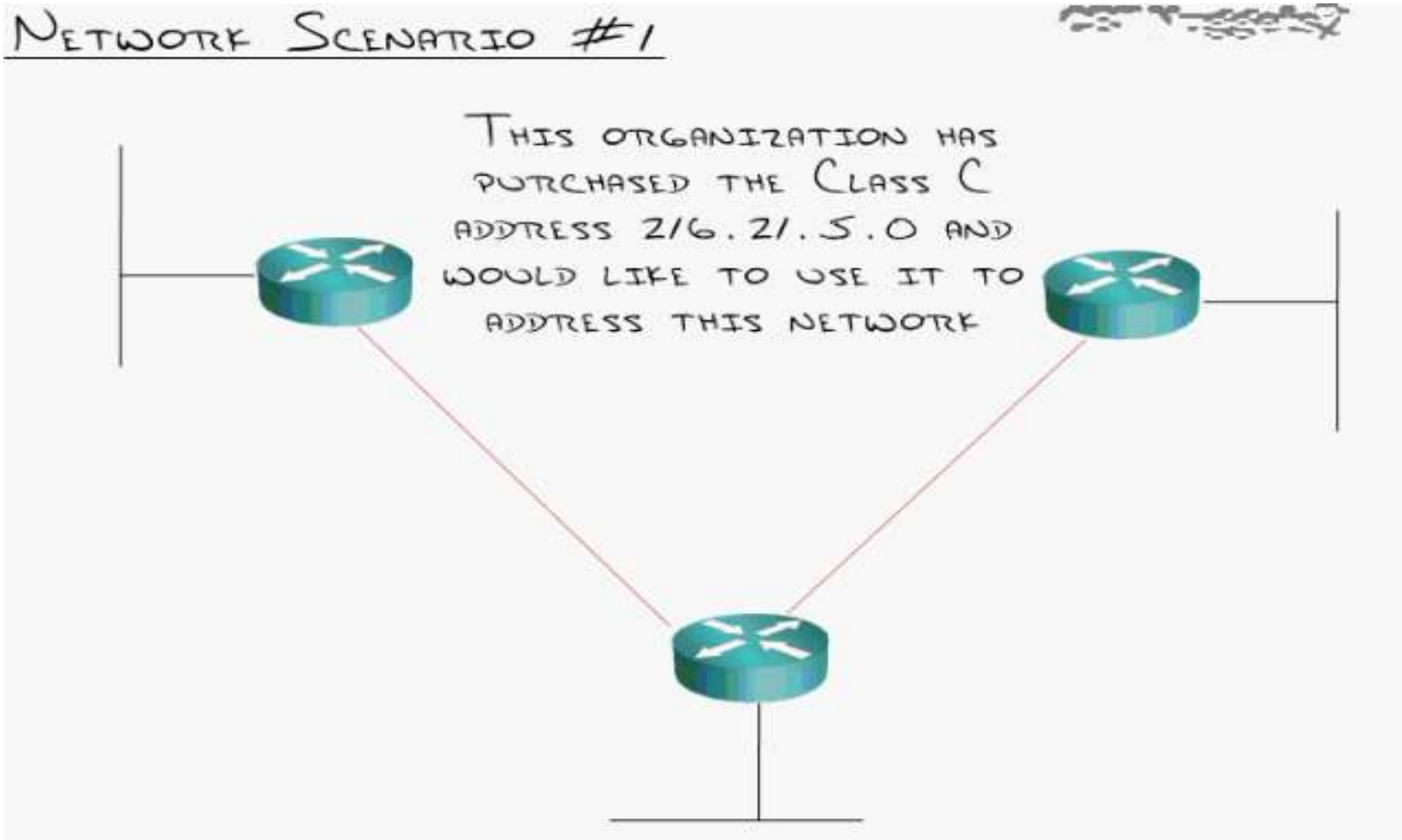
.

-IANA is an organisation which lists all registered port numbers. All the above listed port numbers can be found on ***www.iana.org/assignments/port-numbers*.**

***Note:*** In order to see what is source and destination port numbers from your machine, visit different websites and type ***C:\>netstat –f or netstat –n .***

***Note: The tale of packets was explained in the previous slides***

# Advanced TCP/IP: Subnetting based on Number of Networks.



# NETWORK SCENARIO #1

216.21.5.0

255.255.255.0

1 = NET

HOST = 0

1. DETERMINE NUMBER OF NETWORKS AND CONVERT TO BINARY

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |

5 = 00000101

3 BITS

2. RESERVE BITS IN SUBNET MASK AND FIND YOUR INCREMENT

255.255.255.0 = 11111111.11111111.11111111.00000000

255 . 255 . 255 . 224 = 11100000

128
+64
+32
224

32

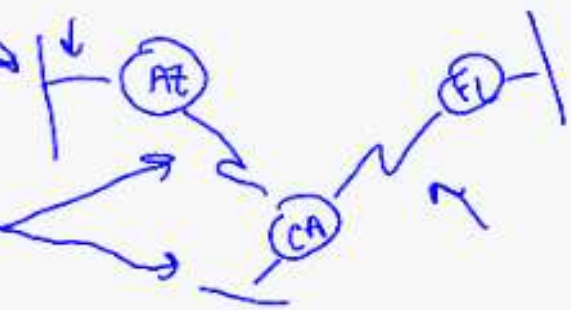
3. USE INCREMENT TO FIND YOUR NETWORK RANGES

216.21.5.0 – 216.21.5.31

216.21.5.32 – 216.21.5.63

216.21.5.64 – 216.21.5.95

216.21.5.96



# NETWORK SCENARIO #2

CLASS [illegible] 20.0

NEED: 50 NETWORKS

$\begin{array}{r} 250 \\ -32 \\ \hline 18 \end{array}$

$\begin{array}{r} 255 \\ -3 \\ \hline 252 \end{array}$

1. DETERMINE NUMBER OF NETWORKS AND CONVERT TO BINARY

1 1 1 1 1 1 00

128 64 32 16 8 4 2 1

0 0 1 1 0 0 1 0

50 = 00110010

6 BITS

2. RESERVE BITS IN SUBNET MASK AND FIND YOUR INCREMENT

255.255.255.0 = 11111111.11111111.11111111.00000000

255.255.255.252 1111 1100

4

3. USE INCREMENT TO FIND YOUR NETWORK RANGES

195.5.20.0 1-2 195.5.20.3

195.5.20.4 5-6 195.5.20.7

195.5.20.8 9-10 11

.20.12 15

.20.16 19

# NETWORK SCENARIO #3

150.5.0.0
255.255.0.0

CLASS B: 150.5.0.0

NEED: 100 NETWORKS

1. DETERMINE NUMBER OF NETWORKS AND CONVERT TO BINARY

128 64 32 16 8 4 2 1

0 1 1 0 0 1 0 0

100 = 01100100

7 BITS

2. RESERVE BITS IN SUBNET MASK AND FIND YOUR INCREMENT

255.255.0.0 = 11111111.11111111.00000000.00000000

/16 = /23

255. 255. 11111110. 00000000

254 0

2

150.5.2.0

3. USE INCREMENT TO FIND YOUR NETWORK RANGES

- 1 150.5.0.0 – 150.5.1.255
- 2 150.5.2.0 – 150.5.3.255
- 3 150.5.4.0 – 150.5.5.255
- 4 150.5.6.0
  150.5.8.0

150.5.0.255
150.5.1.0 } VALID

$2^{n-7}$

$\lfloor 2^{n-7} \rfloor$

# NETWORK SCENARIO #4

CLASS A: [illegible].0.0

NEED: 500 NETWORKS

500
-256

1. DETERMINE NUMBER OF NETWORKS AND CONVERT TO BINARY

SR 256 128 64 32 16 8 4 2 1

1

500 - 9 BITS

$(2^{n-8})$

2. RESERVE BITS IN SUBNET MASK AND FIND YOUR INCREMENT

255.0.0.0 = 11111111.00000000.00000000.00000000

11111111.10000000.

128

255.255.128.0

/17

128
+128
256

3. USE INCREMENT TO FIND YOUR NETWORK RANGES

10.0.0.0 - 10.0.127.255
10.0.128.0 - 10.0.255.255
10.1.0.0 - 10.1.127.255
10.1.128.0 - 10.1.255.255
10.2.0.0

# Subnetting based on number of Hosts.

NETWORK SCENARIO #1

30 HOSTS

216.21.5.0
255.255.255.0

1. DETERMINE NUMBER OF HOSTS AND CONVERT TO BINARY

128 64 32 16 8 4 2 1
0 0 0 1 1 1 1 0

30 = 00011110

5 BITS

2. RESERVE BITS IN SUBNET MASK AND FIND YOUR INCREMENT

255.255.255.0 = 11111111.11111111.11111111.00000000

/27

255.255.255.224 — 11100000

32

$2^5$

3. USE INCREMENT TO FIND YOUR NETWORK RANGES

216.21.5.0 – 31 (1-30)
.5.32 – 63
.5.64 – 95
.96 – 127
.128
.160

$(2^3) = 8$

# NETWORK SCENARIO #2

CLASS [illegible] 20.0

NEED: 50 HOSTS PER NETWORK

1. DETERMINE NUMBER OF HOSTS AND CONVERT TO BINARY

| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | | | | | |

50 = 6 BITS

2. RESERVE BITS IN SUBNET MASK AND FIND YOUR INCREMENT

255.255.255.0 = 1111...111.00000000

11000000

/26 .192 64 $(2^6) - 2 = 62$ HOSTS

3. USE INCREMENT TO FIND YOUR NETWORK RANGES

195.5.20.0 - 63

.20.64 - 127

.128 - 191

.192 - 255

~~256~~

$(2^2) = 4$

# NETWORK SCENARIO #3

CLASS [illegible].0.0

NEED: 500 HOSTS PER NETWORK

1. DETERMINE NUMBER OF HOSTS AND CONVERT TO BINARY

512 [256 128 64 32 16 8 4 2 1]
1

500 = 9 BITS

2. RESERVE BITS IN SUBNET MASK AND FIND YOUR INCREMENT

255.255.0.0 ⇒ 111...11.00000000.00000000

255.255.254.0 .11111110.00000000

/23

2

3. USE INCREMENT TO FIND YOUR NETWORK RANGES

150.5.0.0 – 150.5.1.255
150.5.2.0 – 150.5.3.255
.5.4.0
.6.0
.8.0
⋮

# NETWORK SCENARIO #4

CLASS A: [illegible].0.0

NEED: 100 HOSTS PER NETWORK

1. DETERMINE NUMBER OF HOSTS AND CONVERT TO BINARY

128 | 64 32 16 8 4 2 1

1 →

100 = 7 BITS

2. RESERVE BITS IN SUBNET MASK AND FIND YOUR INCREMENT

255.0.0.0 = 111...00000000.00000000.00000000

255.255.255.128 ← 11111111 10000000 → 128

3. USE INCREMENT TO FIND YOUR NETWORK RANGES

10.0.0.0 – 127

10.0.0.128 – 255

10.0.1.0 – 1.127

1.128 – 1.255

2.0

2.128

$2^{\wedge 17}$

# Reverse Engineering Subnet.

REVERSE ENGINEERING, SCENARIO #1

IP: 192.168.1.127

MASK: 255.255.255.224

128 64 32 16 8 4 2 1

1 1 1 00000

Ans

11100000

32

192.168.1.0

.32

.64

192.168.1 .96 – 127

.128

# Reverse Engineering, Scenario #2

IP: 172.16.68.65

MASK: 255.255.255.240

GW: 172.16.68.62

IP: 172.16.68.62

MASK: 255.255.255.240

128 64 32 16 8 4 2 1

1 1 1 1 0 0 0 0

172.16.0.0

16

11110000

172.16.1

172.16.68.0

.16

.32

.48-63

172.16.68.64-79

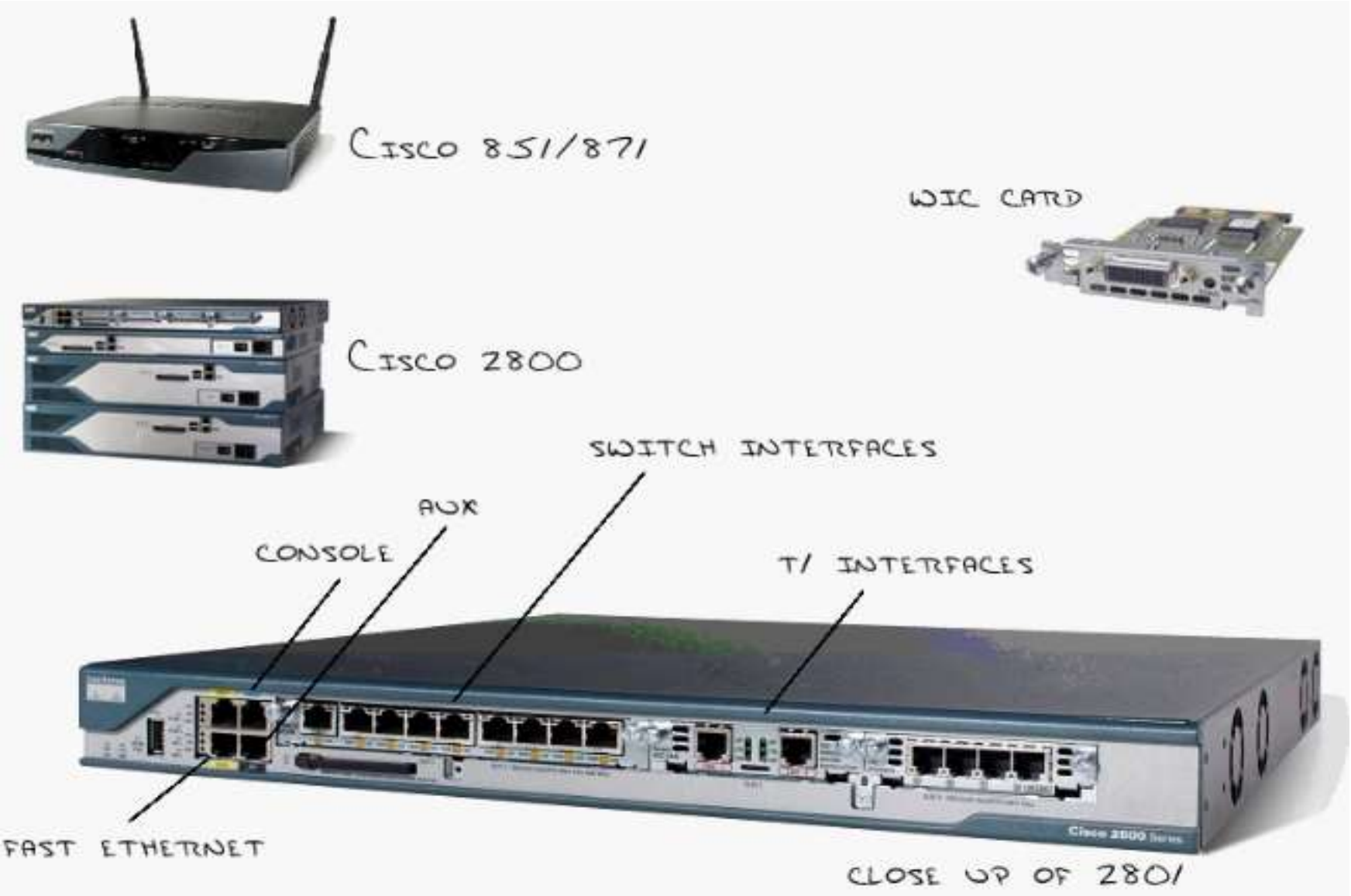
.80

2.0

# Initial setup of Cisco Switch / Router.

- Understanding physical indicators.
- Observing the booting process.
- Performing the initial configuration.

# Understanding physical indicators.



# Observing the booting process.

```
System Bootstrap, Version 12.3(8r)YI, RELEASE SOFTWARE
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2005 by cisco Systems, Inc.

C850 series (Board ID: 4-149) platform with 65536 Kbytes of main memory

Booting flash:/c850-advsecurityk9-mz.123-8.YI2.bin
Self decompressing the image : ########################################################### [
OK]

              Restricted Rights Legend

Use, duplication, or disclosure by the Government is
subject to restrictions as set forth in subparagraph
(c) of the Commercial Computer Software - Restricted
Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
Software clause at DFARS sec. 252.227-7013.

           cisco Systems, Inc.
           170 West Tasman Drive
           San Jose, California 95134-1706


Cisco IOS Software, C850 Software (C850-ADVSECURITYK9-M), Version 12.3(8)YI2, RELEASE SO
FTWARE (fc1)
Synched to technology version 12.3(10.3)T2
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Tue 14-Jun-05 20:08 by ealyon
Image text-base: 0x8002008C, data-base: 0x80F991F8
```

.

- The IOS version of the router is ***12.3(8r)YI.***
- Inside the booting flash, that is where the IOS image is.***c850-advsecurityk9-mz.123-8.YI2.bin***
- It has **128K bytes** of Non-volatile RAM which is used to store the configurations.
- It has as well **20480K bytes** of processor board system **flash** which is used to store the IOS.
- ***Would like to enter the initial configuration dialog?[yes/no]:no.***

# Basic Switch Configurations

- ***Switch>***

- This is user mode

- ***Switch > enable*** or ***en***

- This will take you to privilege exec mode.

- ***Switch #***

- This is privilege mode.

- ***Switch # Configure terminal*** or ***config t*** or ***conf t***

- This will take you to global configuration mode.

- ***Switch(config)#***

- This is global configuration mode.
- This where you can make changes in your router or switch.

# Basic Switch Configurations

- ***Switch(config)# hostname SW***

*- This command used to change the name of the switch*

- ***SW(config)#***
- *SW(config)#banner motd @Unauthorized access to this switch is illegal.@*

- After doing any changes on a switch or router save the configurations in the NVRAM by using the command ***copy running-config startup-config***
- *Use the **show running-config** command to view the current configuration.*
- *Use the **show startup-config** command to view the configuration saved in NVRAM*

# Basic Switch Configurations

- Configuring the line console will secure your router. Setting password to your switch or router.

*SW(config)# line console 0*

*SW(config-line)# password cisco*

*SW(config-line)# login*

*SW(config-line)# exec-timeout 30 0*

*SW(config-line)# exit*

*SW(config)# line vty 0 4*

*SW(config-line)# password cisco*

*SW(config-line)# login*

*SW(config)# enable password cisco*

*SW(config)# enable secret ciscos*

*SW(config)# exit*

*SW # copy run start*

*SW # reload*

# VLAN

- A Virtual LAN (VLAN) is simply a logical LAN, just as its name suggests.
- VLANs have similar characteristics with those of physical LANs, only that with VLANs, you can logically group hosts even if they are physically located on separate LAN segments.
- We treat each VLAN as a separate subnet or broadcast domain.
- For this reason, to move packets from one VLAN to another, we have to use a **router** or a **layer 3 switch**.

# VLAN configurations

- VLANs are configured on switches by placing some interfaces into one broadcast domain and some interfaces into another.
- An ***access port*** is assigned to a single VLAN . These ports are configured for switch ports that connect to devices with a normal network card, for example a PC in a network.
- A ***trunk port*** on the other hand is a port that can be connected to another switch or router. This port can carry traffic of multiple

*Switch#config terminal*

*Switch(config)#vlan 10*

*Switch(config-vlan)#name SALES*

*Switch(config-vlan)#vlan 20*

*Switch(config-vlan)#name IT*

*Switch(config)#int fa0/1*

*Switch(config-if)#switchport mode access*

*Switch(config-if)#switchport access vlan 10*

*-- do this for every interface*

# Inter-VLAN Routing

- Enable communication between the two VLANs via a single physical interface.
- It is done on the router by dividing the single physical interface on the router into logical interfaces (sub interfaces).
- Each sub-interface will then serve as a default gateway for each of the VLANs.
- This scenario is called **Router on a Stick** (ROAS) and will allow the VLANs to communicate through the single physical interface.

# Inter-VLAN Routing

- The "**encapsulation dot1q**" command in networking configures a sub-interface to handle IEEE 802.1Q VLAN tagging, enabling it to receive and transmit packets with a specified VLAN ID.
- This command is used on Layer 2 sub-interfaces to associate them with a particular VLAN, effectively creating a trunk link for VLAN traffic
- *Eg. Router(config-subif)# encapsulation dot1q <VLAN ID>*

# Inter-VLAN Routing

- *Router> enable*
- *Router# config terminal*
- *Router(config)# int fa0/0*
- *Router(config-if)# no shutdown*
- *Router(config-if)# int fa0/0.10*
- *Router(config-subif)# encapsulation dot1q 10*
- *Router(config-subif)# ip add 192.168.1.1 255.255.255.0*
- *Router(config-subif)#*
- *Router(config-subif)# int fa0/0.20*
- *Router(config-subif)# encapsulation dot1q 20*
- *Router(config-subif)# ip add 192.168.2.1 255.255.255.0*

# Enable Telnet on a Cisco Switch or Router

- **Telnet,** an acronym for “Teletype Network” is a network protocol that enables users to remotely access and control another computer or device over a network.
- Telnet's primary disadvantage is its lack of security.
- All data transmitted through Telnet, including usernames, passwords, and commands, is sent in plain text, making it vulnerable to interception and unauthorized access.
- This lack of encryption makes Telnet unsuitable for sensitive communications and has largely been replaced by more secure protocols like SSH

# Enable Telnet on a Cisco Switch or Router

To enable telnet, you need to do two basic things;

- Set up Management IP address which is the default VLAN of the switch or into the interface of the router to connect to.
- Set up the remote VTY channels, then enable the telnet

# Enable Telnet on a Cisco Switch or Router

- *Switch(config)# interface vlan 1*

For router, put the interface of the router

- *Switch(config-if)# ip address <IP>  <mask>*
- *Switch(config-if)# no shutdown*
- *Switch(config)# line vty 0 15*
- *Switch(config-line)# transport input telnet*
- *Switch(config-line)# password <password>*
- *Switch(config-line)# login*

# Enable Telnet on a Cisco Switch or Router

- From computer terminal execute the command *telnet <IP address>* to remotely connect the switch or router.
- You can also telnet from switch to the router and vice versa.
- For you to be able to do configuration from telnet (get privilege mode), you must set the enable password or secret.

# Enable SSH on a Cisco Switch or Router

- SSH is the abbreviation for Secure Socket Shell or Secure Shell.
- Unlike Telnet, SSH is a protocol to connect to a remote server or system securely.
- It uses an SSH (a secure protocol) as the data transfers between the client and the host in an encrypted manner.

# Enable SSH on a Cisco Switch

To configure SSH on a Cisco switch, you need to follow the following steps;

- Assign an IP address to the default VLAN,
- Generate RSA keys,
- Set a hostname and domain name,
- Create user accounts, and
- Enable SSH on the VTY lines.

# Enable SSH on a Cisco Switch

- *Switch(config)# hostname S1*
- *S1(config)# banner motd 'This is Cisco Switch'*
- *S1(config)# interface vlan 1*
- *S1(config-if)# ip address <IP> <mask>*
- *S1(config-if)# no shutdown*
- *S1(config-line)# exit*
- *S1(config)# ip default-gateway <IP on the router interface>*

- *S1(config)# enable secret <secret>*
- *S1(config)# line console 0*
- *S1(config-line)# password <password>*
- *S1(config-line)# exit*
- *S1(config)# service password-encryption*

# Enable SSH on a Cisco Switch

- *S1(config)# username <username> secret <secret>*
- *S1(config)# ip domain-name admin*
- *S1(config)# crypto key generate rsa*

The name for the keys will be: S1.admin

How many bits in the modulus [512]: **1024** (Recommended)

- *S1(config)# ip ssh version 2*
- *S1(config)# line vty 0 15*
- *S1(config-line)# transport input ssh*
- *S1(config-line)# login local (*enables local login authentication*)*

# Enable SSH on a Cisco Switch

- From a computer, use the ssh command to connect to the switch using the configured IP address and username.
- For example: ssh -l <username> <router or switch_ip_address>.

# Enable SSH on a Router

To enable SSH on the router is same as that of Switch, except that you need to configure the IP address to the interface of the Router.

- *Router(config)# hostname R1*
- *R1(config)# banner motd 'This is Cisco Router'*
- *R1(config)# interface gig0/0*
- *R1(config-if)# ip address 192.168.1.1 255.255.255.0*
- *R1(config-if)# no shutdown*
- *R1(config-if)# exit*
- *R1(config)# ip domain-name <domain-name>*
- *R1(config)# username <username> secret <password>*
- *R1(config)# enable secret cisco*

# Enable SSH on a Router

- *R1(config)# line console 0*
- *R1(config-line)# password cisco*
- *R1(config-line)# login*
- *R1(config-line)# exit*
- *R1(config)# service password-encryption*
- *R1(config)# crypto key generate rsa*

*How many bits in the modulus [512]: 1024*

- *R1(config)# ip ssh version 2*
- *R1(config)# line vty 0 15*
- *R1(config-line)# transport input ssh*
- *R1(config-line)# login local*

# Configuring Switch Port Security

- Switch port security helps to restrict unauthorized devices from accessing the network through that port.
- To configure port security on a Cisco switch, you enable the feature on a specific interface, set the maximum number of secure MAC addresses allowed, and optionally configure the violation mode.

# Configuring Switch Port Security

- Navigate to the specific interface you want to configure e.g *fa0/1*
- Make sure the interface is configured as an access port using *switchport mode access* command.
- Enable port security on the interface using the command: *switchport port-security*
- Specify the maximum number of MAC addresses allowed on the port. The default is 1.

Use the command: *switchport port-security maximum <number of allowed MAC addresses>*

# Configuring Switch Port Security

- Configure the violation mode, use the command: *switchport port-security violation <mode>*
- Cisco switches offer different violation modes to handle unauthorized access attempts:
  - **protect**: Drops packets from unknown MAC addresses, but the port remains up. No SNMP trap is sent.
  - **restrict**: Drops packets and sends an SNMP trap or log message indicating a violation. The port stays up.
  - **shutdown**: The default action, which shuts down the port and sends an SNMP trap or log message.

# Configuring Switch Port Security

- SNMP traps are asynchronous messages sent from an SNMP-enabled device to a management system (the "manager") to notify it of events or significant status changes on the device, without being explicitly requested.
- Essentially, they are unsolicited alerts that provide real-time information about what's happening on the network.

# Configuring Switch Port Security

- Specify the allowed MAC address to access that interface using the command: *switchport port-security mac-address <MAC address(es)>,* or
- Enable sticky MAC address learning to dynamically learn MAC addresses and add them to the running configuration.

Use the command: *switchport port-security mac-address sticky*

- To view the port security status and configurations, use the command *show port-security interface <interface>*
- To view all secure MAC addresses on the switch, use the command *show port-security address*

# Configuring Switch Port Security

- *Switch(config)# interface range <allowed ports>*
- *Switch(config-if-range)# switchport mode access*
- *Switch(config-if-range)# switchport port-security maximum <number of allowed MAC>*
- *Switch(config-if-range)# switchport port-security mac-address sticky*
- *Switch(config-if-range)# switchport port-security violation restrict*
- *Switch(config-if-range)# exit*
- *Switch(config)# interface range <remaining ports>*
- *Switch(config-if-range)# shutdown*
- *Switch# show port-security interface <interface>*
- *Switch# show port-security address*

# IP Routing

- IP routing is the process by which routers select the optimal path for data packets to travel from their source to their destination on a network.
- It's a fundamental aspect of data communication, especially on the Internet, where networks are interconnected.
- Routers are devices that perform IP routing.
- They maintain routing tables that contain information about network paths.
- IP routing operates at the network layer (Layer 3) of the OSI model.

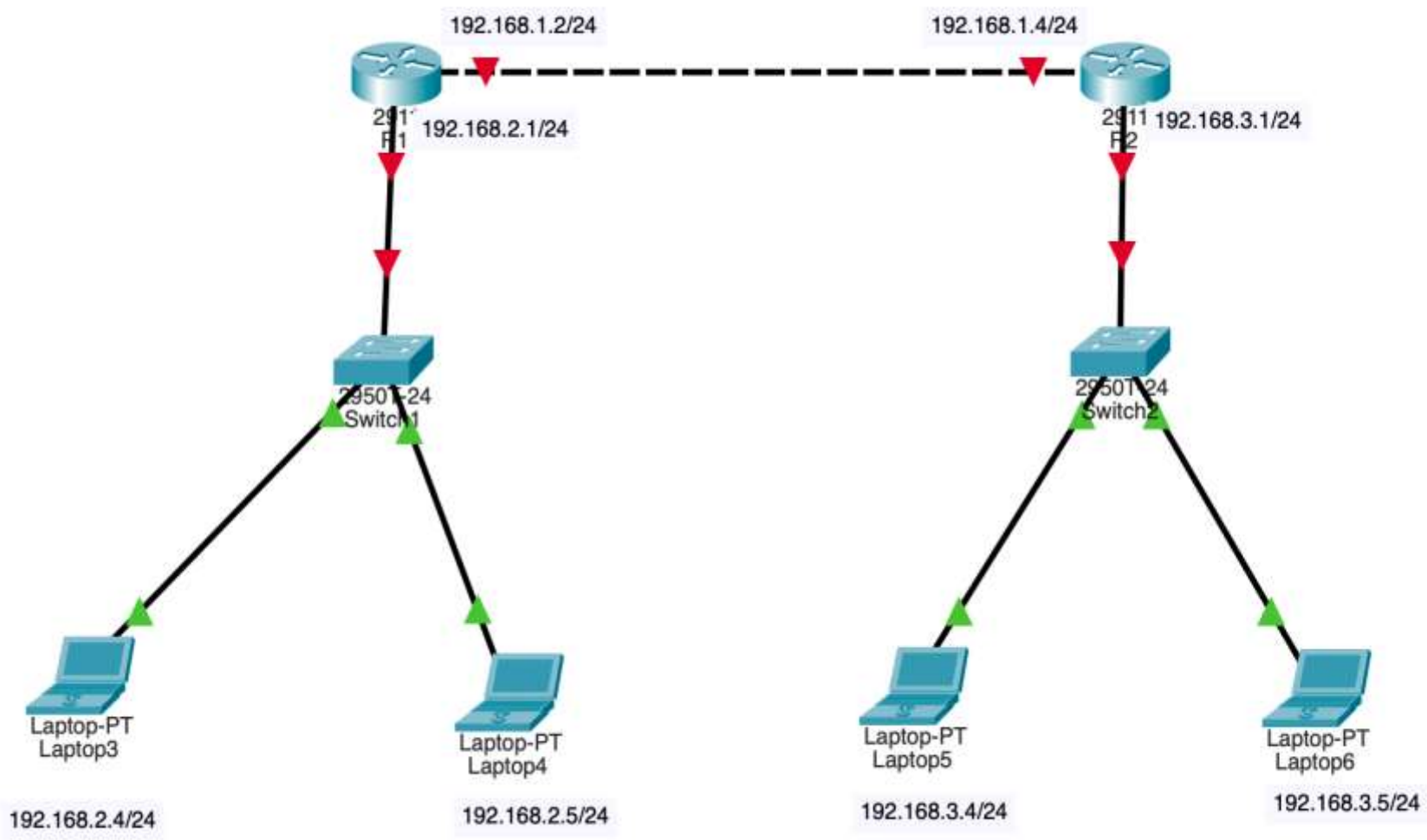
# IP Routing

- When a router receives a packet, it examines the destination IP address and uses the routing table to determine the next hop (the next router or destination) in the path.
- The routing table provides the necessary information for the router to select the best path for the packet to reach its destination.
- The **Routing Protocols** are used to setup the IP routes.
- Routing Protocols are protocols used to dynamically exchange routing information between routers, helping them build and maintain their routing tables.

# IP Routing

- **Static routing:** network administrators manually configure routing entries in the router's routing table.
- **Dynamic routing:** routers learn routing information from other routers using routing protocols.
- *Advantages and disadvantages of static routing?*
- *Advantages and disadvantages of dynamic routing?*

# Static Routing



# Static Routing

- Assign an IP address and subnet mask using the command *ip address <ip_address> <subnet_mask>* to every interface.
- Activate the interface with *no shutdown* command.
- Use the command *ip route <destination_network> <subnet_mask> <next_hop_ip_address>* to define a static route.
- Use the *show ip route* command on each router to verify that the static routes have been added to the routing table.

# Dynamic Routing Protocols

- Dynamic routing uses a route that a network routing protocol adjusts automatically for topology or traffic changes.
- When the router is turned on it will send routing information to other neighbour routers telling them what it knows, allowing the router to build paths automatically.
- Once the router learn about the new network it will keep the information in the **routing table.**

# Dynamic Routing Protocols

- Dynamic routing is further categorized to **distance vector, link state** and **hybrid**
- Distance vector protocols rely on exchanging routing tables with neighbours.
- Link-state protocols flood the network with link-state information to build a complete network map
- This leads to differences in convergence speed, scalability, and resource utilization
- *What are the advantages and disadvantages of distance vector and link state protocols?*

# Dynamic Routing Protocols

- **Distance vector.**

- Easy to configure.
- It is slow to converge especially when one of its interface goes down.
- It doesn’t have many features.
- RIP, IGRP and BGP.

# Dynamic Routing Protocols

- **Link state.**

- Difficult to configure(more knowledge required).
- Feature-riffic
- OSPF, IS-IS.

# Dynamic Routing Protocols

- **Hybrid.**

- The best of both world.
- It has both features of Distance vector and Link state.
- Proprietary- it works with only cisco devices
- EIGRP.

# Routing Information Protocol (RIP)

- A relatively simple distance-vector protocol, often used in smaller networks or for basic routing tasks
- Its algorithm first developed in 1969.
- It comes in two version: RIPv1 and RIPv2.

**RIPv1.**

-Classful version (Does not support VLSM) –It doesn't send the subnet mask when sends its routing information to neighbours.

-No authentication- You can just bring a router and plug in.

-It uses broadcast – Sends broadcast every after 30sec.

- It has a maximum of 15 hop counts

# Routing Information Protocol (RIP)

**RIPv2.**

- Classless version (Support VLSM) – It send its routing information update with subnet mask information.
- Adds MD5 authentication.
- It uses multicast 224.0.0.9 –send information to a specific group.
- It has a maximum of 15 hop counts

# Routing Information Protocol (RIP)

- Enter global configuration mode
- Start RIP routing process by using the command *router rip*
- Specify RIP version (optional): *version 2* for RIPv2, which supports CIDR and VLSM.
- Use the *network* command followed by the *network address* of each directly connected network to advertise the networks. E.g *network <network address>*
- Repeat the process on all routers in your topology.

# Routing Information Protocol (RIP)

- It is recommended to finish the configuration by using *no auto-summary* command.
- The primary purpose of the no auto-summary command is to disable the automatic route summarization that some dynamic routing protocols perform.
- When routes are summarized automatically, only the network portions of the addresses are advertised to other routers, which can lead to issues in networks that use subnetting.

# Routing Information Protocol (RIP)

- For example, if you have a number of subnets (like 192.168.1.0/24 and 192.168.2.0/24), RIPv1 would automatically summarize these to a classful address (192.168.0.0).
- By using *no auto-summary*, you can advertise the specific subnets instead.
- RIPv2 does not automatically summarize routes, as it supports Classless Inter-Domain Routing (CIDR). Thus, using *no auto-summary* in RIPv2 is generally unnecessary.
- Use the command *show ip route* on each router to see the learned routes (routing tables).

# Open Shortest Path First (OSPF)

- A widely used link-state protocol known for its scalability and efficiency, commonly deployed within large organizations or enterprises.
- To configure OSPF in Cisco Packet Tracer, you need to enable OSPF on each router, define the networks and areas, and optionally configure router IDs and other settings.
- Router ID is configured using the command *router-id <router_id>*
- If router ID is not configured, the router will automatically choose the highest IP address on an active interface as its router ID.

# Open Shortest Path First (OSPF)

- Enable OSPF using the command *router ospf <process_id>*
- The process ID is a locally significant number (1-65535) that differentiates multiple OSPF processes on a router.
- The process ID should be the same across all the routes in the network.
- Define OSPF networks and areas using the command *network <network_address> <wildcard_mask> area <area_number>*
- The area number specifies the OSPF area the network belongs to.
- You can configure multiple networks and areas on a single router.

# Open Shortest Path First (OSPF)

- Wildcard mask is the flipped version of the subnet mask.
- The wildcard mask indicates which portions of the IP address should be matched (0 = match, 1 = ignore).
- To determine a wildcard mask, you typically subtract the subnet mask from 255.255.255.255.
- For example, for a /24 network (255.255.255.0), the wildcard mask would be 0.0.0.255
- For a /30 network (255.255.255.252), the wildcard mask would be 0.0.0.3

# Open Shortest Path First (OSPF)

- Alternatively, you can advertise the specific interface instead of advertising the network address and wildcard mask
- Enter to the specific interface
- Use the command *ip ospf <process-id> area <area-number>*
- Do that for all interfaces which are directly connected to the router.

# Enhanced Interior Gateway Routing Protocol (EIGRP)

- A Cisco-proprietary **hybrid** protocol that combines features of both distance-vector and link-state protocols.
- It's known for its convergence speed and support for complex network topologies.
- To configure EIGRP in Cisco Packet Tracer, you need to enable the EIGRP routing process on the routers, specify the directly connected networks, and configure passive interfaces.
- Disable automatic summarization if the network topology contains discontiguous networks.

# Enhanced Interior Gateway Routing Protocol (EIGRP)

- Enable EIGRP: *router eigrp <Autonomous System number>*
- Advertise directly connected networks using the command: *network <network address> <wildcard mask>*
- Configure passive interfaces to prevent EIGRP updates on specific interfaces: *passive-interface <interface>*
- ✓ The passive-interface command within an EIGRP configuration prevents the router from sending or receiving EIGRP routing updates on a specific interface
- ✓ The passive interface is configured to the interfaces connected to networks where dynamic routing isn't needed, such as LANs connected to a switch.

# Enhanced Interior Gateway Routing Protocol (EIGRP)

- Disable automatic summarization: *no auto-summary*
- Check neighbor adjacencies: *show ip eigrp neighbors*
- Display EIGRP routing information: *show ip route eigrp*

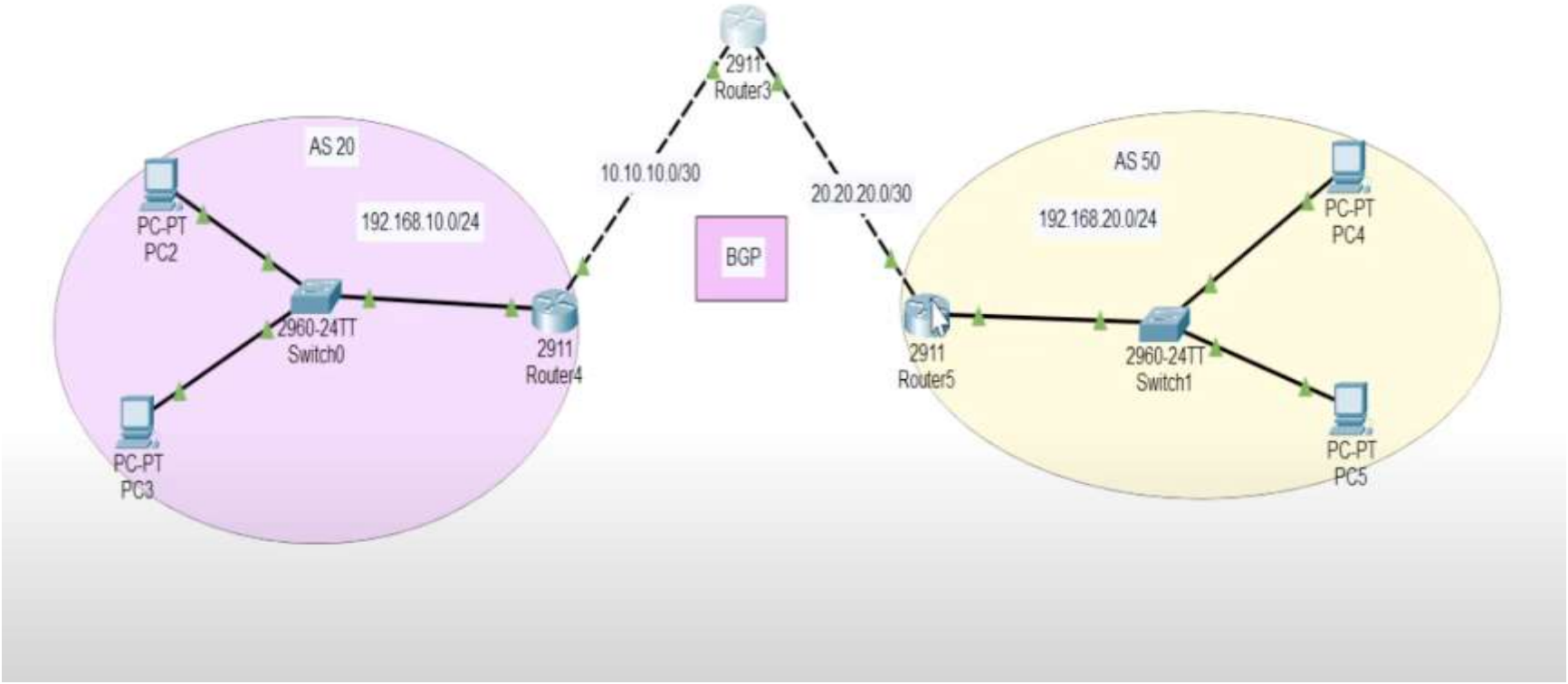
# Border Gateway Protocol (BGP)

- The de facto standard for routing between different autonomous systems (AS) and the internet.
- It uses path-vector routing and is crucial for internet connectivity.
- Configure BGP on each router, defining the autonomous system number (ASN) and establishing neighbour relationships and advertise the directly connected networks.

# Border Gateway Protocol (BGP)

- Enable the BGP routing process using the command *router bgp <ASN>*, where <ASN> is the autonomous system number.
- Set the router ID with the command *bgp router-id <router_ID>*
- Establish a BGP neighbour relationship with other routers using the command *neighbor <neighbor_IP> remote-as <neighbor_ASN>*
- Advertise the networks connected to the router using the command *network <network_address> mask <subnet_mask>*
- Use the command *show ip bgp summary* to verify the BGP neighbour status.
- Use the command *show ip bgp* to view the BGP routing table.

# Border Gateway Protocol (BGP)



# Access Control Lists (ACLs)

- An Access Control List (ACL) is a set of rules that govern network traffic, dictating which users or systems can access specific resources.
- These rules are essentially filters applied to network devices like routers, switches, or firewalls, determining whether to allow or deny traffic based on various criteria.
- ACLs enhance network security by preventing unauthorized access and controlling the flow of data within a network.
- ACLs consist of a series of rules that define permitted and denied traffic.

# Access Control Lists (ACLs)

- They act as filters, examining incoming and outgoing network traffic based on defined criteria.
- By controlling access, ACLs help protect sensitive resources from unauthorized access and potential threats.
- ACLs are commonly implemented on routers, switches, and firewalls, but can also be found in operating systems.
- ACLs can filter traffic based on various criteria, such as IP addresses (source and destination), protocols, ports, and other parameters.

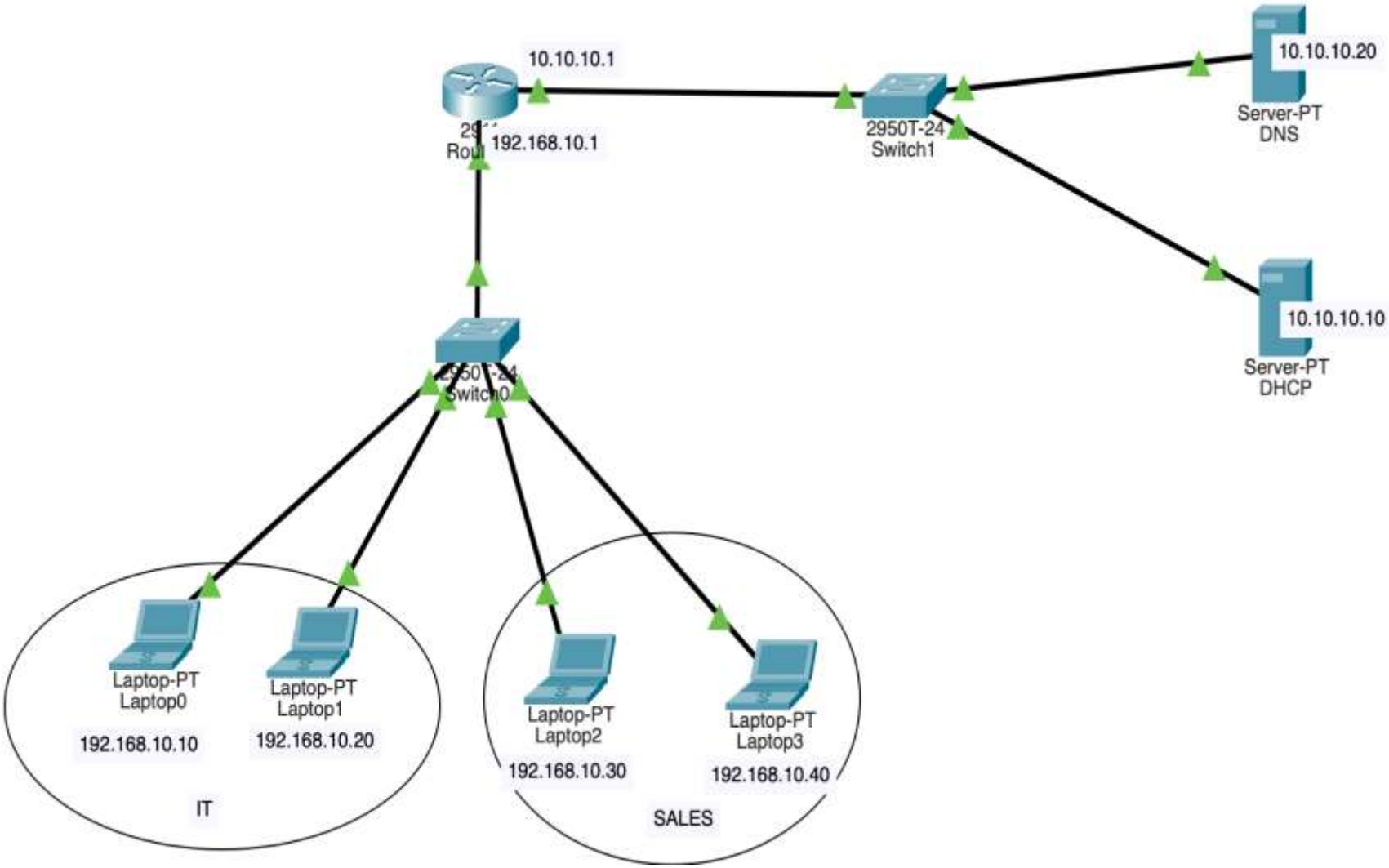
# Access Control Lists (ACLs)

- If a packet doesn't match any rule in the ACL, it is typically denied by default.
- ACLs can be implemented for both inbound and outbound traffic on network interfaces.
- There are various types of ACLs, including standard, extended, dynamic, and reflexive, each with different rule application capabilities

# Access Control Lists (ACLs)

- Create the ACL using the access-list command, followed by a number (1-99 for standard, 100-199 for extended) and either permit or deny followed by the traffic criteria (source IP for standard, source/destination IP, protocol, etc. for extended).
- Finally, apply the ACL to a specific interface in the desired direction (in or out) using the *ip access-group* command.

# Access Control Lists (ACLs)



# Standard ACLs (1-99)

- Use the command *access-list <access-list-number> <permit|deny> <host|source source-wildcard|any>*.
- For example, to permit traffic from IT hosts to access the servers :

✓ *access-list 10 permit host 192.168.10.10*

✓ *access-list 10 permit host 192.168.10.20*

- To deny all other traffic: *access-list 10 deny any*

# Extended ACLs (100-199)

- Use the command *access-list <access-list-number> <permit|deny> <protocol> <source> <destination> [options]*
- For example, to permit the traffic from IT to access only the DHCP server (10.10.10.10)
- *access-list 110 permit ip 192.168.10.10 255.255.255.0 10.10.10.10 255.255.255.0*
- *access-list 110 permit ip 192.168.10.20 255.255.255.0 10.10.10.10 255.255.255.0*
- Deny any other traffic to access DHCP server using*: access-list 110 deny ip any any*

# Extended ACLs (100-199)

- Use the command *access-list <access-list-number> <permit|deny> <protocol> <source> <destination> [options]*
- For example, to deny TCP traffic from a specific network to a specific port on another network: *access-list 101 deny tcp 192.168.1.0 0.0.0.255 host 10.0.0.10 eq 80*

# Applying the ACL to an Interface

- Enter to the specific interface
- Use the command *ip access-group <access-list-number> <in|out>* to apply the ACL.
- For example, to apply ACL 10 in the incoming direction: *ip access-group 10 in*
- Assume you want block all traffic from the 10.0.0.0/24 network from reaching a server connected to the GigabitEthernet 0/1 interface.

# Applying the ACL to an Interface

- Assume you want block all traffic from the 10.0.0.0/24 network from reaching a server connected to the GigabitEthernet 0/1 interface.
- Create a standard ACL: *access-list 10 deny 10.0.0.0 0.0.0.255*
- Permit all other traffic: *access-list 10 permit any*
- Enter interface configuration mode for Gig 0/1: *interface gig 0/1*
- Apply the ACL in the incoming direction: *ip access-group 10 in*

# Configuring ACL to the VTY Lines

- Suppose you want only the PCs from IT department to be able to remotely access the router;
- Configure remote access on the router and test SSH on both departments.
- Configure a standard ACL to only permit the two IT PCs to remotely access the router.
- Bind the ACL created on the VTY interfaces using the command: *access-class <ACL number> in*